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An Improved Cuckoo Search Algorithm for System Efficiency in Cloud Computing

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Abstract

In the last few years, cloud computing is a model which is enabled for on demand and suitable system access to the computing resources. Cloud computing services through the IoT (Internet of Things). Many advance optimization algorithms are used in different environments. In this paper, we only focused on Cuckoo Search Algorithm (CS), this algorithm reduced the execution time and optimize the Virtual Machine. Cuckoo Search (CS) algorithm to improve the system efficiency. The proposed approach is calculating the fitness value by using MATLAB framework, this algorithm is produce the better performance in the system.

Key Words: IoT, Cuckoo Search, Virtual Machine

1. Introduction

Cloud computing is used to model the scientific application in the field of astronomy, bioinformatics and physics. It is an effective and promising computing model. Different virtual Machines (VM) vary the computation capacity because of slow and old machines and it is continuously replaced by a new and fast machines. Hence heterogeneous cloud computing systems are special systems in Virtual Machine (VM).

Computational Intelligence [1] describes to simulate the virtual inspections of CYP2A6, BIRC3 and improve the system efficiency. Xi'an Jiao tong [2] preserved a data analysis method to collect data, process, analyze and visualize collected data. Map reduces and optimization technique demonstrates the use of cloud computing technique and analyzes the weather data. There are two cases first technique is mainly focus on the forecasting temperature and the second cases is the map reduce technique. By using the above technique execution time in the cloud is reduced and finally achieves the data visualization. Mofijul Islam, Abdur Razzaque et al. [9] propose a Genetic Algorithm (GA) based Virtual Machine (VM) technique in heterogeneous mobile cloud computing to reduce the execution time. Here, to optimize the effectiveness of the VM migration user mobility and load of the cloud servers is taken into consideration. The main aim is to reduce the total number of Virtual Machine (VM) migration and select the optimal cloud server. Elina Pacini et al. [10] preserved a Particle Swam Optimization (PSO) technique to increase the system efficiency and it is compared with random assignment and Genetic algorithms. Ashraf Darwish et al. [3] proposed an Internet of Things (IOT) and cloud computing to solve problems in healthcare applications such as remote medical services, smart hospitals and medicine control. Here it produce the new concept for CC and IOT health paradigm and its main aim is to present the state of art ,gap analysis between for different levels of integration components. Ahmed Abdelaziz, Mohamed Elhoseny et al. [4] proposed new intelligent algorithms such as Genetic Algorithm (GA), Parallel Particle Swam optimization (PPSO) and Particle Swam Optimization (PSO) to find the optimal chosen of the Virtual Machine (VM) in the cloud environment. To find the optimal chosen in cloud computing cloud sim is used and to find the

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optimal intelligent algorithm matlab tool is used. Hence it is proved that PPSO is a better optimization technique compared to GA and PSO algorithm. The organization of the paper is described as follows: Section 2 briefly describes the system model of the proposed technique and cuckoo search optimization technique. Experimental result and analysis is discussed in section 3 and section 4 draws the conclusion based on the experimental results.

2. Method

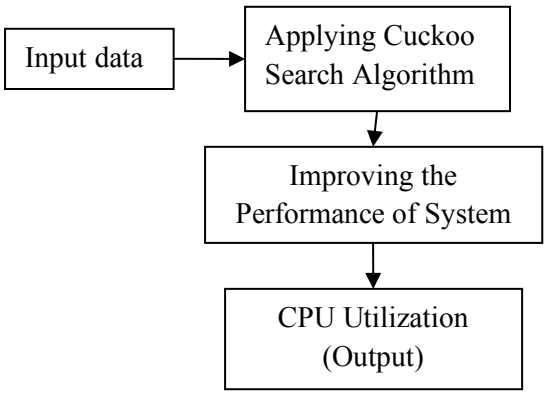


Figure: 1 Block Diagram of the proposed system

Figure.1, initially, input data is given to the system. After applying CS algorithm, the input values are optimized to improve the performance of the system. Finally, the elapsed execution time of the system proved that the Cuckoo Search (CS) algorithm is better.

Cuckoo Search Algorithm

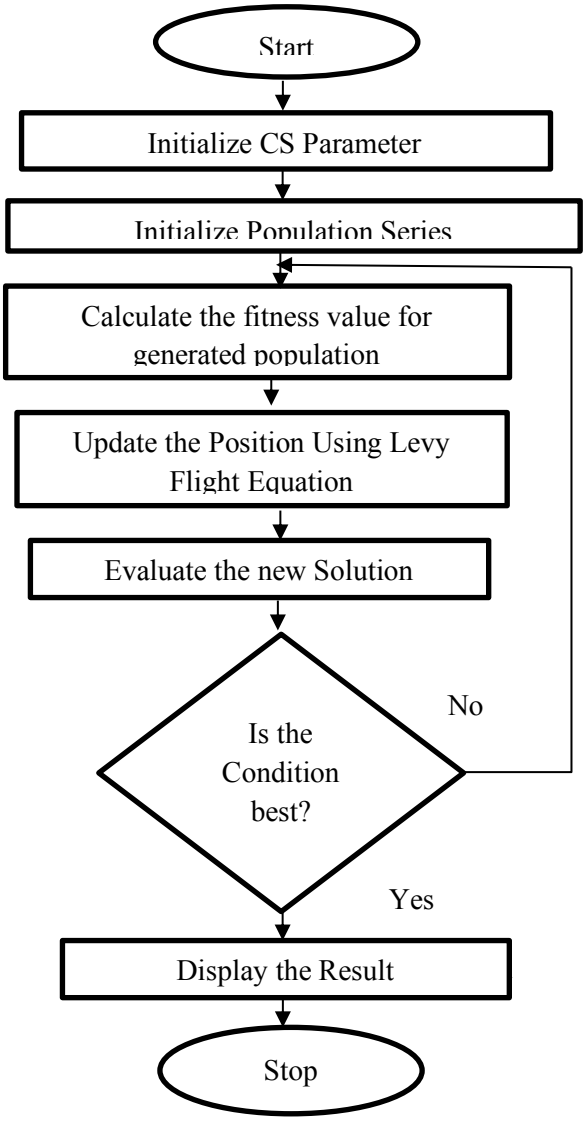
Yang and Deb propose a Cuckoo Search (CS) algorithm in the year 2009 and it is a heuristic algorithm. The parasitic parent and levy flight of the cuckoo are simulated to solve the optimization problem in Cuckoo Search (CS) algorithm.

Cuckoo does not need nest during breeding because it laid its own eggs in other nest and the other birds reproduce the eggs. The birds find hatching and breeding by its own and it spawn the eggs. When the bird is out cuckoo allows the bird to replace the off spring.

In order to balance the local and global optimization technique the random walk generates the length of the short and long steps. Random walk is levy flight and it simplifies the cuckoo parasitism in nature.

Steps:

1. At a time a cuckoo lays one egg and it chooses a parasitic nest for hatching purpose based on random walk technique.
2. Only the best nest is used for next generation based on the selected parasitic bird nest.
3. If the number of nest is parasitic, the owner of nest finds the probability of foreign eggs and chooses to discard the eggs.



In Cuckoo Search (CS) algorithm, the parameter pa is controlled by the combination of global and local explorative random walk. The local random walk equation is shown below,

$$x_i^{t+1} = x_i^t + s \otimes H(pa - \varepsilon) \otimes \left(x_j^t - x_k^t\right)$$

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Randomly select the two different solutions x_j^t *and* x_k^t based on random permutation Heaviside function is expressed as $H(u)$. From uniform distribution a random number is drawn and it is denoted by ε, s is the step size. The global random walk is carried out by using Levy flights

$$x_i^{t+1} = x_i^t + \alpha L(s, \lambda)$$

Where

$$L(s, \lambda) = \frac{\lambda \Gamma(\lambda) \sin(\pi\lambda/2)}{\pi} \quad s > s_0 > 0$$

Algorithm

f(x) is the Objective function and $x=(x_1, \ldots\ldots\ldots x_d)^T$
Generate initial population of n host nests x_i
While (t<Max Generation) or (stop criterion)
Randomly get a cuckoo or generate a solution by Levy flights
And then evaluate the quality or fitness F_i
Choose a nest among n(say, j) randomly
if$(F_i > F_j)$,
Replace j by the new solution
End
Abandon a fraction (pa) of worse nests & generate new solutions
Keep best solutions (or nests with quality solutions)
Rank the solution and find the current best
End while
Post process result and visualization

3. Experimental Analysis

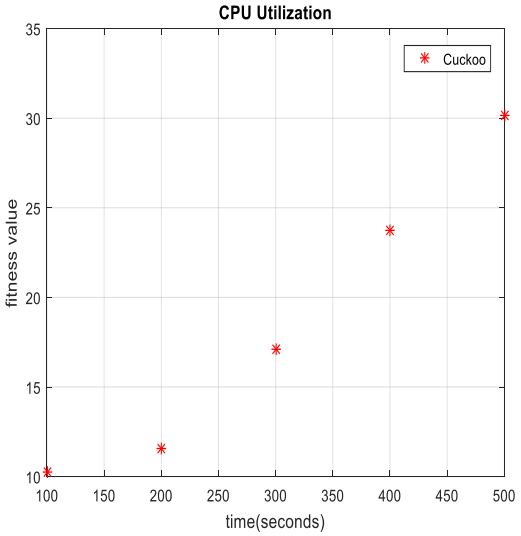


Figure: 3 CPU Utilization result of CS Algorithm

Cuckoo search algorithm the execution time increases by increasing the fitness value. The maximum calculated time is 500 sec and obtained fitness value for 500sec is 30.5

Conclusion

For the past few year, cloud computing is considered as the emerging technology. And execution time of the system is concerned as the main issue. This paper mainly focused on optimizing the execution time and increasing the performance of the system. CS Algorithm is the optimization algorithm performed on the system to optimize the CPU utilization thereby reducing the execution time. It is proved that the CS algorithm provides better results than existing methods.

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